

Bridging Theory and Practice in Machine Learning Education Through Mixed Reality Gamification

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Abstract—Teaching the abstract principles of machine learning (ML) remains a considerable challenge within computer science education. This study presents IMREG, an immersive mixed-reality (MR) educational game, coupled with a new framework that uses gamification to improve the understanding of core ML topics such as image recognition, reinforcement learning, and linear regression. The IMREG includes game-based learning in an MR setting, allowing students to interactively alter the inputs and parameters of the model. During the first stage, users modify the values of the image processing parameters. In the subsequent stage, they direct the behavior of the agent to influence the dynamics of reinforcement learning. The final stage involves systematically gathering data samples, visualizing outcomes through charts, and modifying regression coefficients to see prediction updates in real time. Engagement is increased through intuitive natural interactions, including gesture detection and eye-tracking. A quantitative study with 20 undergraduate students from both computer science and other disciplines reported positive responses regarding usability and engagement. A post-intervention assessment evaluated the knowledge of the three ML concepts. The participants answered three sets of multiple-choice questions, achieving a mean accuracy of 76.83%. In particular, students from non-computer science backgrounds obtained higher scores (mean score of 81.39%) than their computer science counterparts (mean score of 70%), suggesting the IMREG's capacity to widen access to ML education. These results support an effective MR game design framework, offering valuable guidance for creating inventive and captivating teaching approaches in computer science.

Index Terms—mixed reality, education, serious game, gamification

I. INTRODUCTION

Computer science education faces persistent challenges in fostering deep understanding and sustained engagement, particularly for complex topics like machine learning (ML). Although educational games show promise for hands-on learning [1], it remains difficult to effectively integrate entertainment with substantive educational value into higher education curricula [2]. Successful educational game design requires careful attention to user engagement, intuitive usability, and measurable learning outcomes [3], [4]. Mixed reality (MR) emerges as a transformative tool for immersive learning, allowing real-time interaction with physical and virtual elements while

minimizing the risks of cybersickness inherent in immersive environments [5]. Although MR has proven effective in fields such as medical training [6], its potential for computer science education, particularly in simplifying abstract ML concepts, remains underexplored.

Current research recognizes the benefits of combining MR with gamification and serious games in computer science education [7]. However, few studies investigate the role of MR in bridging the gap between theoretical knowledge of ML and practical application [8]. This gap highlights the need for interactive tools that engage students with basic concepts of ML more effectively than traditional methods, which often fail to stimulate interest or connect theory to real-world scenarios [9]. Conventional approaches, which are based heavily on lectures and theoretical exercises, also lack scalability and adaptability, further limiting engagement and retention [10].

Wadsworth [11] and Long [12] demonstrated that students often report feeling overwhelmed by the disconnect between theory and practice. Furthermore, resource limitations, including insufficient access to computational tools and relevant datasets, prevent institutions from providing effective ML education [13], [14]. Karalekas *et al.* [15] endorse STEM-based robotics as an effective approach to teaching ML concepts in K-12 education. Hernandez *et al.* [16], however, report substantial resource requirements, including specialized equipment and expert instruction [17] for similar engineering projects in complex educational contexts. These demands potentially restrict the broader implementation within higher ML education.

Addressing these challenges requires innovative educational approaches. Project-based learning, gamification, and constructivist methods show considerable promise to improve student engagement and connect theoretical knowledge with practical application [18], [19]. These approaches increase student motivation and provide immediate feedback, promoting a deeper understanding of complex ML processes [20]. Studies indicate that MR environments, by blending real and virtual elements, can bridge the theory-practice gap. MR makes complex ML concepts more accessible and comprehensible, offering a promising approach to hands-on learning experiences [21].

This paper presents IMREG, an MR-based educational

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game designed to teach core ML concepts, including image recognition, reinforcement learning, and linear regression. IMREG uses gamification to create an interactive learning environment in which students manipulate the inputs and parameters of the ML model in real time. By including natural interactions such as gesture detection and eye tracking, the system promotes intuitive comprehension and engagement. The open exploration space aims to deepen conceptual understanding, addressing the limitations of traditional ML pedagogy [22]. Gamification principles, particularly when combined with cooperative problem-solving, have demonstrated significant improvements in engagement and retention [23], [24]. In line with constructivist learning theory [25], IMREG promotes active knowledge construction through environmental interaction.

This study addresses the following research questions (RQs):

- RQ1: How can MR-based educational games utilize innovative design to improve both conceptual understanding and engagement in introductory ML?
- RQ2: What design principles for integrating gamification and educational content in MR games significantly influence user experience and learning outcomes for complex ML concepts?

The study evaluates IMREG through user experience analysis and feedback, employing established questionnaires to iteratively refine the game and its underlying design framework. The research also proposes a novel framework that balances engagement with educational effectiveness, ensuring enjoyable, yet durable learning. Specifically, it investigates how multi-modal interactions, such as gaze tracking, hand detection, and real-time model manipulation, improve immersion, interactivity, and learning outcomes. In addition, the study explores the optimal balance of entertainment and education to maximize engagement and knowledge acquisition.

II. METHODOLOGY

This study employs a design-based research approach to develop and evaluate IMREG, enabling iterative refinement through user feedback and performance data. This methodology ensures a balance between engagement and educational effectiveness, aligning with established educational game design frameworks. The approach is particularly suited for optimizing educational interventions in real-world contexts while contributing to theoretical advancements. IMREG structures gameplay around core ML stages, including data collection, pre-processing, feature selection, model training, and evaluation, embedded within a narrative framework that transforms abstract concepts into interactive tasks, thereby reinforcing comprehension and practical relevance.

The methodology focuses on two critical components, including (1) the systematic application of game design principles, combining pedagogical frameworks and narrative integration to meet research objectives; and (2) the technical implementation of MR, emphasizing innovative interaction modalities to enhance immersion and usability.

A. Game Design Principles

The IMREG design integrates established educational research models to optimize learning outcomes and user engagement. Based on constructivist learning theory, the game implements three key frameworks: TRACK [26] for content structure, RECIPE [27] for engagement evaluation, and MEEGA+ [28] for game mechanics assessment. The design extends these frameworks by adapting narrative elements and challenges to utilize MR's unique capabilities for teaching abstract ML concepts.

The game experience lasts approximately 40 minutes, with flexible interaction options (sitting or standing) to promote natural movement and comfort in the MR environment. Figure 1 illustrates the core design principles, which combine constructivist learning foundations, experiential learning approaches, and educational game design frameworks.

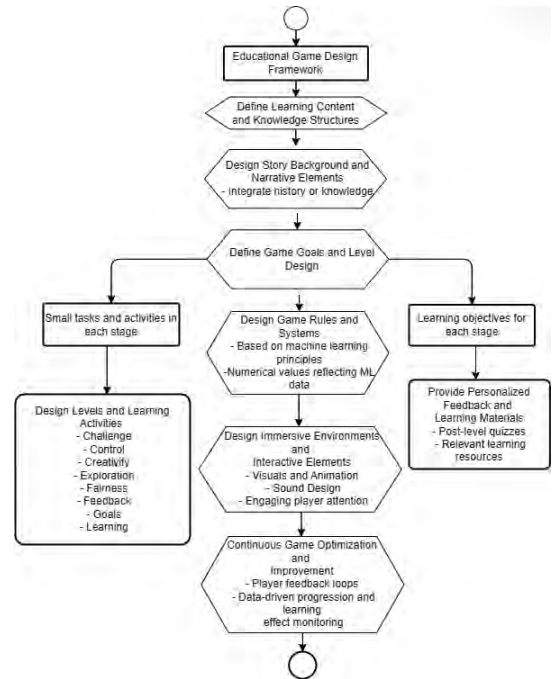


Fig. 1: IMREG's game design principles framework, showing integration of constructivist learning theory [25], experiential learning, and educational game design models.

B. Game Mechanisms

IMREG employs a narrative-driven framework that seamlessly blends interactive gameplay with ML education, transforming abstract concepts into tangible applications. The players assume the role of a spaceship pilot who collaborates with a virtual assistant (Athena) to discover habitable planets. The linear progression through three key stages, illustrated in Fig. 2, ensures immediate application of learned concepts:

1) *Stage 1 - Identity Verification (Image Recognition)*: This stage simulates image recognition workflows, including image pre-processing, feature extraction, and facial recognition, via interactive controls to solidify conceptual understanding. The



Fig. 2: IMREG's three-stage structure: (1) identity verification (image recognition), (2) spaceship piloting (reinforcement learning), and (3) planetary exploration (linear regression), with each stage building upon previous concepts through narrative continuity.

interface allows users to modify image processing parameters, for instance, by cropping image dimensions, to receive immediate visual feedback on how different pre-processing techniques affect the outcome. This experiential method transforms abstract ideas into concrete, interactive experiences, overcoming the drawbacks of traditional passive instruction. Fig. 3 illustrates this process.

2) *Stage 2 - Spaceship Piloting (Reinforcement Learning)*: Following successful identity verification, Athena guides players to pilot the spaceship by providing mission objectives and control instructions. Then, the players take control of the spaceship, followed by the explanations of reinforcement learning principles from Athena, with a focus on reward mechanisms. Players navigate to the target planet, passing through checkpoints to receive rewards and strategically avoiding obstacles. This simulation introduces key reinforcement learning concepts such as agents, rewards, objectives, and the underlying principles of Q-Learning in an experiential manner, providing a practical understanding that complements theoretical knowledge. Fig. 4 illustrates this stage.

3) *Stage 3 - Planetary Exploration (Linear Regression)*: The players teleport to a planetary environment where safety is maintained by visually highlighting real-world obstacles. Guided by the character Athena, users undertake data collection through a shooting simulation, proceed to data processing through visualization interactions, and apply these steps to complete equations. A key activity involves manipulating the slope and intercept of the regression line to observe how these adjustments alter the prediction outcomes in real time. This stage explains linear regression by grounding its principles in direct visual and interactive experimentation. Fig. 5 shows this stage.

C. Interaction Methods

IMREG uses MR's spatial capabilities to create immersive learning environments through two primary settings: a spaceship control room and planetary surface. The system overlays virtual interfaces onto physical spaces using pass-through technology, enhancing presence while reducing cybersickness risks. The planetary environment employs safety boundaries



(a)



(b)



(c)

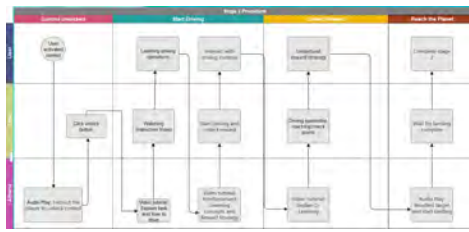
Fig. 3: Overview of stage 1 components: (a) conceptual workflow, (b) pre-processing interaction, and (c) facial recognition implementation.

and interactive 3D objects with dynamic animations and spatial audio to reinforce realism. The game supports three complementary interaction modalities, accommodating diverse learning preferences and MR experience levels:

- **Gesture Recognition:** Meta Quest 3's hand tracking enables natural manipulation of virtual objects through intuitive gestures, promoting direct engagement with ML concepts. This approach eliminates controller barriers for first-time users.
- **Controller Input:** Traditional controllers provide precision for tasks requiring fine motor control, particularly during the data collection shooting mini-game in Stage 3's linear regression activities.
- **Eye Tracking:** Gaze-based interaction reduces cognitive load by enabling natural focus-driven selection of in-game elements, streamlined the learning process.

D. Participants and Experimental Setup

The study involved 20 Chinese university students (45% female, 50% male, 5% non-disclosed) with minimal prior knowledge of ML. Participants from non-computer science



(a)



(b)



(c)

Fig. 4: Overview of stage 2 components: (a) conceptual framework, (b) navigation interface, and (c) stage transition.

majority represented diverse academic fields, including business, engineering, and social sciences. The demographic data of the participants revealed that 65% of the participants had experience playing with MR. The educational materials were presented in English and the participants self-reported their language proficiency as intermediate (55%), beginner (30%), advanced (10%) or fluent (5%). The university's ethics review board approved the study protocol and all participants gave their informed consent prior to participation.

E. Experiment Procedure

The experiments were carried out in a standardized classroom environment with controlled lighting and minimal distractions. Each participant occupied a dedicated 3×3×4 meter safety zone for unrestricted movement while using the Meta Quest 3 headset. A researcher supervised each session to ensure protocol compliance and participant safety throughout the experiment.

Each 60-minute experimental session followed a structured three-phase protocol:

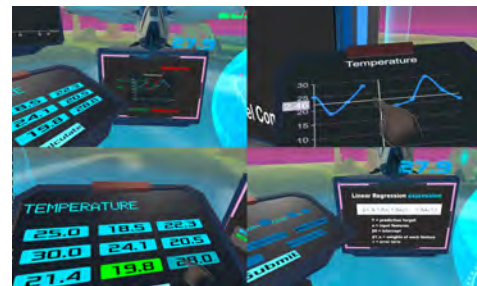
- 1) **Preparation Phase (15 minutes):** System familiarization with MR interaction methods (gestures/controllers)



(a)



(b)



(c)



(d)

Fig. 5: Overview of stage 3 components: (a) conceptual process, (b) data tools, (c) equation completion, and (d) prediction output.

and baseline assessment through the ML knowledge test before the game.

- 2) **Gameplay Phase (30 minutes):** Unassisted completion of the IMREG three-stage ML curriculum and continuous observation by researchers of safety and protocol adherence.
- 3) **Evaluation Phase (15 minutes):** Post-game knowledge assessment, self-rated benchmark questionnaires, and open-ended feedback collection.

Quantitative analysis compared pre- and post-test scores to measure knowledge gain. The standardized questionnaires

evaluated three dimensions: engagement (GEQ, [29]), usability (UEQ, [30]) and educational effectiveness (MEEGA+). Qualitative data from participant feedback and researcher observations provided contextual insights for design improvements.

III. RESULTS

Analysis of pre- and post-test performance and questionnaire responses from all 20 participants demonstrates IMREG's dual effectiveness in both knowledge acquisition and learner engagement. The results reveal three key findings: (1) a significant improvement in the understanding of the ML concept in all participants' groups, (2) consistently high engagement metrics throughout the game, and (3) strong user acceptance of the game-based learning methodology. These outcomes are examined through three aspects that correspond to our research objectives: learning outcomes, user engagement and experience, and usability of the IMREG. Quantitative data are supplemented by qualitative feedback to provide comprehensive information on the educational effectiveness of IMREG.

The differences between the computer science and non-computer science majors were analyzed with independent sample *t*-tests, applying the Mann-Whitney U test as normality was violated. The correlations between self-rated prior knowledge and post-test performance were evaluated using Pearson's *r* and Spearman's ρ . For standardized questionnaires (GEQ, UEQ, MEEGA+), mean scores and standard deviations are reported with single-sample *t*-tests and 95% confidence intervals compared with these results.

A. Learning Outcomes

The pre-game evaluations confirmed limited machine learning familiarity among participants, with a mean score of 2.6 out of 5 ($\sigma = 1.2$). Seventy percent of users (14 participants) reported having a minimal prior understanding of ML. Only five participants demonstrated brief familiarity with the specific concepts covered: image recognition, reinforcement learning, and linear regression. This baseline knowledge level validated the suitability of the participant sample for evaluating beginner-focused educational interventions.

Post-game evaluations revealed significant knowledge gains, as detailed in Table I. All participants achieved scores exceeding 50%, with 70% (14 participants) scoring above 70%. These results demonstrate effective knowledge transfer of foundational ML concepts. Participants with computer science majors obtained a mean score of 70% ($\sigma = 13.80$, $n = 8$), which is significantly less than the mean score of 81.39% ($\sigma = 8.22$, $n = 12$) among those majoring in other disciplines ($t(18) = -2.32$, $p = .032$, $d = 1.06$). This performance pattern suggests particular effectiveness for learners without extensive computer science backgrounds. No significant correlation occurred between self-assessed prior knowledge and final test scores, indicating consistent knowledge acquisition across initial proficiency levels. The potential influence of English proficiency requires consideration, as 85% of the participants self-rated below advanced proficiency.

TABLE I: Participant demographics and performance metrics: ID, English proficiency (1 = beginner to 4 = fluent), academic major, self-reported ML knowledge, and post-test scores

ID	English	Major	ML Knowledge	Test Score
1	1	CS	Very Poor	73.33%
2	1	Non-CS	Very Poor	86.67%
3	2	CS	Poor	73.33%
4	3	CS	Very Familiar	66.67%
5	1	CS	Poor	66.67%
6	2	Non-CS	Fair	93.33%
7	2	CS	Poor	66.67%
8	3	Non-CS	Familiar	73.33%
9	1	Non-CS	Poor	86.67%
10	2	Non-CS	Poor	86.67%
11	2	CS	Familiar	100.00%
12	2	CS	Familiar	53.33%
13	2	Non-CS	Fair	80.00%
14	1	Non-CS	Fair	90.00%
15	1	Non-CS	Very Poor	73.33%
16	2	CS	Fair	60.00%
17	2	Non-CS	Very Poor	66.67%
18	2	Non-CS	Fair	73.33%
19	2	Non-CS	Fair	80.00%
20	4	Non-CS	Fair	86.67%
Mean				76.83%

The MEEGA+ evaluation (Fig. 6) further substantiated positive learning outcomes. The participants expressed a strong agreement on perceived knowledge gains: 55% agreed and 35% strongly agreed that the game improved cognitive understanding of machine learning concepts. Elevated focused attention scores additionally confirmed sustained engagement throughout the learning experience.



Fig. 6: MEEGA+ evaluation results displaying mean scores across eight dimensions: relevance, learning, focused attention, fun, satisfaction, challenge, perceived learning, and social interaction (scale: -2 = strongly disagree to +2 = strongly agree).

B. User Engagement and Experience

The GEQ assessed engagement across multiple dimensions. The results of the core module (Fig. 7a), the in-game module (Fig. 7b) and the post-game module (Fig. 7c) collectively demonstrate consistently positive experiences. The key engagement metrics significantly exceeded the moderate benchmarks: *sensory and imaginative immersion* ($\mu = 3.2$), *imaginative immersion* ($\mu = 3.1$), *flow* ($\mu = 3.0$), and *positive affect* ($\mu = 3.3$) indicate deep engagement of the participants. The *challenge* scores ($\mu = 2.1$) approached the benchmark levels, confirming the appropriate difficulty in balancing sustained engagement without excessive frustration. Elevated *competence* scores ($\mu = 3.0$) reflect the sense of achievement of the participants throughout the game. The minimal *tension* scores ($\mu = 0.8$) and the *negative affect* scores ($\mu = 0.7$) confirm the absence of significant adverse experiences.

The post-game module (Fig. 7c) indicated a generally positive overall experience, with a high score for positive experience. The social presence module (Fig. 7d) revealed strong *empathy* ($\mu = 3.4$) and *behavioral involvement* ($\mu = 3.2$), further validating the immersive qualities of the game. The post-game evaluations showed high positive experience scores ($\mu = 3.5$) with rapid transitions from reality to reality ($\mu = 2.1$), indicating effective immersion without disorientation.

C. System Usability

UEQ results (Fig. 8a) consistently demonstrate positive user feedback across all dimensions. The participants in particular highly rated experiential aspects, including enjoyable, understandable, pleasant, and motivating. The benchmark comparison with the official UEQ data ((Fig. 8b) vs. Fig. 8c) confirms above average performance in all six usability categories: *attractiveness*, *perspicuity*, *efficiency*, *dependability*, *stimulation*, and *novelty*. *Attractiveness* and ($\mu = 2.3$) and *stimulation* ($\mu = 2.1$) achieved an “excellent” rating, indicating strong visual appeal and participation.

The MEEGA+ usability assessment (Fig. 9) further validates these findings. The participants were awarded high scores for *aesthetics* ($\mu = 1.8$), confirming the effective visual design in the layout, graphics and color scheme of the interface. Elevated *learnability* scores ($\mu = 1.7$) and *operability* ratings ($\mu = 1.6$) indicate intuitive understanding of rules despite the limited MR experience of the participants.

Taken together, these results establish IMREG as an effective educational tool with high usability, particularly beneficial for learners without extensive computing backgrounds. The IMREG delivers a consistently positive user experience while effectively conveying the foundational concepts of ML.

IV. DISCUSSION

A. Addressing RQ1

IMREG integrates narrative, interaction, and game mechanics. This integration enables students to intuitively learn and apply key ML concepts, including image recognition, reinforcement learning, and linear regression. The results of the post-game test demonstrate the effectiveness of IMREG,

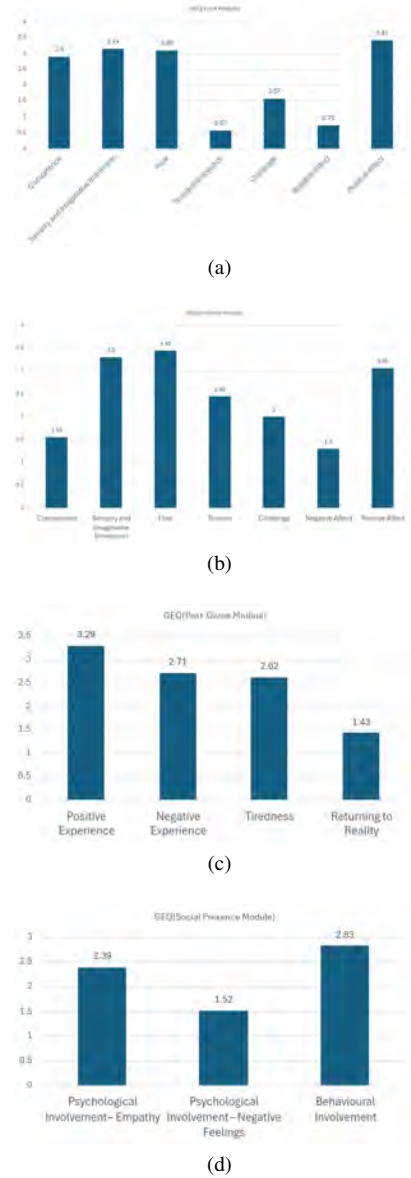


Fig. 7: GEQ results across four assessment modules (scale: 0 = not at all to 4 = extremely) which include (a) core, (b) in-game, (c) post-game and (d) social presence modules. Error bars represent standard deviations.

with participants achieving a significant mean score of 76.83%. These results illustrate how MR’s unique advantages, such as immersive environments and spatial interaction, combined with engaging game design principles facilitate deep conceptual understanding and sustain student engagement.

In particular, non-computer science students outperformed their computer science counterparts. This finding suggests that IMREG’s design effectively lowers barriers to entry for learners without specialized backgrounds, highlighting the potential of MR games to democratize access to complex technical concepts. Furthermore, the results of the questionnaire show that 95% of the participants expressed positive

bridging theoretical understanding and practical application. IMREG uses MR's unique capabilities to create diverse interaction opportunities that facilitate deep conceptual comprehension while enhancing learner engagement. The results demonstrate that well-designed MR educational games significantly improve learning outcomes and engagement, offering innovative pathways for teaching complex technical subjects.

The study acknowledges two primary limitations: First, the 20-participant sample from a single university limits generalizability to broader populations and diverse educational contexts. Second, participants' mainly below-average self-reported English proficiency may have influenced game comprehension and performance metrics. Future work will expand validation through larger and more diverse participant cohorts in multiple institutions. Furthermore, integrating dynamic difficulty adjustment mechanisms could further personalize the learning experience by automatically adapting challenge levels to individual learner progression.

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